THE CLOW–ZAGUIA COP BOUND IS TIGHT AT INDEPENDENCE NUMBER FOUR: $f(4) = 3$

Abstract. Clow and Zaguia proved that every connected perfect graph G with $\alpha(G) \geq 4$ satisfies $\mathrm{c}(G) < \alpha(G)$, and asked for the least function f with $\mathrm{c}(G) \leq f(\alpha(G))$ on connected perfect graphs. We settle the value $f(4) = 3$. The witness is the complement of the 4×4 rook's graph, equivalently the complement of $L(K_{4,4})$, equivalently the tensor product $K_4 \times K_4$: a 9-regular graph on 16 vertices that is perfect with $\alpha = \theta = 4$ and cop number exactly 3. Both bounds are proved by hand on the printed object; the lower bound $\mathrm{c} \geq 3$ is an explicit robber strategy against two cops, given as a six-case invariant. The graph and its cop number are already in print, in work of Ahirwar, Bonato, Gittins, Huang, Marbach and Zaidman on Latin square graphs; perfection together with $\alpha = 4$ identifies it as a witness for $f(4)$.

1. The problem

We use the game of Cops and Robbers in the convention of Clow and Zaguia [3]: the cops occupy vertices first, the robber then chooses a vertex, and thereafter the two sides move alternately with the cops moving first, each token either staying where it is or moving to an adjacent vertex. The cops win if some cop ever occupies the robber's vertex. The cop number $\mathrm{c}(G)$ is the least number of cops with a winning strategy. As usual $\alpha(G)$ is the independence number and $\theta(G)$ the clique cover number. *Perfect* is used in the form Clow and Zaguia state [3, §1], which by the Strong Perfect Graph Theorem of Chudnovsky, Robertson, Seymour and Thomas [2] is equivalent to the usual one: G contains no induced odd hole C_{2k+1} and no induced odd antihole, $k \geq 2$.

The following is Problem 5.2 of [3], quoted from the arXiv version of that paper.

Problem 5.2 (Clow and Zaguia). Given an integer k, what is the least integer $f(k)$ such that every connected perfect graph G with $\alpha(G) = k$, has $\mathrm{c}(G) \leq f(k)$?

A witness must satisfy all three clauses, and all three are checked below: G connected, G perfect, and $\alpha(G)$ equal to k *exactly*, not merely at least k.

The upper half of the case $k = 4$ is a corollary of the same paper, which proves:

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> If G is a connected perfect graph with $\alpha(G) \geq 4$, then $\mathrm{c}(G) < \alpha(G)$.

Hence $f(k) \leq k-1$ for every $k \geq 4$, and in particular $f(4) \leq 3$. What the $k = 4$ instance of Problem 5.2 therefore needs is a single connected perfect graph with $\alpha = 4$ and cop number 3. We exhibit one, so the bound of [3] is tight at $k = 4$, and $f(4) = 3$.

2. The witness

Let G be the graph whose vertices are the 16 cells (i,j) of a 4×4 grid, $i, j \in \{1,2,3,4\}$, with

$$(i,j) \sim (k,l) \quad \Longleftrightarrow \quad i \neq k \ \text{ and } \ j \neq l. \tag{1}$$

Say two distinct cells *share a line* if they agree in their first or in their second coordinate. Then (1) says exactly that two distinct cells are non-adjacent if and only if they share a line, which is the only fact about G used below. Equivalently G is the complement of the 4×4 rook's graph $K_4 \,\square\, K_4 = L(K_{4,4})$, and also the tensor (categorical) product $K_4 \times K_4$.

The non-neighbours of (i,j) are the other three cells of row i and the other three cells of column j, so G is 9-regular with $|E(G)| = \frac{16 \cdot 9}{2} = 72 = \binom{16}{2} - 48$, the 48 non-edges being the 4 rows and 4 columns at $\binom{4}{2} = 6$ pairs each. Indexing the cell (i,j) as vertex $4(i-1) + (j-1)$ and listing cells in lexicographic order $11, 12, 13, 14, 21, \ldots, 44$, the `graph6` string of G is

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The proofs below check two further objects, printed here.

- The four *broken diagonals* $Q_t = \{(i,\, 1 + ((i-1+t) \bmod 4)) : 1 \leq i \leq 4\}$ for $t = 0,1,2,3$, that is

$$Q_0 = \{(1,1),(2,2),(3,3),(4,4)\}, \quad Q_1 = \{(1,2),(2,3),(3,4),(4,1)\},$$
$$Q_2 = \{(1,3),(2,4),(3,1),(4,2)\}, \quad Q_3 = \{(1,4),(2,1),(3,2),(4,3)\}.$$

- The three-cell set $D = \{(1,1),(2,2),(3,3)\}$.

Lemma 1. *G is connected, perfect, and $\alpha(G) = \theta(G) = 4$.*

Proof. Connected. Two cells sharing a line have a common neighbour, namely any cell off both of their rows and both of their columns; two cells occupy at most 2 rows and at most 2 columns, so at least $4 - 2 = 2$ rows and at least 2 columns remain and such a cell exists.

Perfect. The rook's graph $K_4 \square K_4$ is the line graph $L(K_{4,4})$ of a bipartite graph — the cell (i,j) corresponds to the edge of $K_{4,4}$ joining the ith vertex of one side to the jth vertex of the other, and two cells share a line precisely when the two edges share an endpoint. Line graphs of bipartite graphs are perfect, by König's edge colouring theorem; and the complement of a perfect graph is perfect, by the Perfect Graph Theorem of Lovász [4], which [3, §1] also quotes. Hence $G = \overline{L(K_{4,4})}$ is perfect.

$\alpha = 4$. An independent set S of G is a set of cells that pairwise share a line. Suppose $a, b \in S$ lie in the same row i with distinct columns $j_1 \neq j_2$. Any $c \in S$ off row i must share a line with both, hence must lie in column j_1 and in column j_2, which is impossible. So S lies inside a single row; symmetrically, if two elements of S share a column then S lies inside a single column. Either way $|S| \leq 4$. A full row is independent, so $\alpha(G) = 4$.

$\theta = 4$. Each Q_t has all four rows distinct and all four columns distinct, so by (1) it is a clique, indeed a K_4; and Q_0, Q_1, Q_2, Q_3 partition the 16 cells. Hence $\theta \leq 4$. Also $\theta \geq \alpha = 4$, so $\theta = 4$. □

Lemma 2. $\mathrm{c}(G) \leq 3$.

Proof. $D = \{(1,1),(2,2),(3,3)\}$ is a dominating set: a cell (i,j) undominated by D would have to satisfy $i = 1$ or $j = 1$, and $i = 2$ or $j = 2$, and $i = 3$ or $j = 3$; since i equals at most one of $1, 2, 3$, the value j would have to equal two distinct elements of $\{1,2,3\}$. Three cops placed on a dominating set capture on their first move, so $\mathrm{c}(G) \leq \gamma(G) \leq 3$. □

The lower bound is the substance. It is a strategy for the robber against two cops, kept honest by a single invariant.

Lemma 3. *Let $m \geq 4$ and let $G_m = K_m \times K_m$, that is, the graph on the m^2 cells of an $m \times m$ grid with the adjacency rule* (1)*. Then two cops do not suffice to catch the robber on G_m; hence* $\mathrm{c}(G_m) \geq 3$*. In particular* $\mathrm{c}(G) \geq 3$.

Proof. Call a cell r *safe* for a pair of cop positions (u_1, u_2) if r shares a line with u_1 and with u_2. By (1) this says exactly that $r \notin \{u_1, u_2\}$ and r is adjacent to neither cop. We prove three statements; together they say that the robber can occupy a safe cell after every one of his moves, and therefore is never caught.

(C) Opening. For every pair $(u_1, u_2) = ((a,b),(c,d))$ of cop starting cells, equal cells allowed, a safe cell exists, and the following rule names one. If $a = c$, take $r = (a, e)$ with $e \notin \{b, d\}$, which exists since $m \geq 3 > |\{b,d\}|$; then r lies in row a with both cops and equals neither. Otherwise, if $b = d$, take $r = (e, b)$ with $e \notin \{a, c\}$, symmetrically. Otherwise $a \neq c$ and $b \neq d$; take $r = (a, d)$, which shares row a with u_1 and column d with u_2, and differs from u_1 in the second coordinate and from u_2 in the first.

(A) A cop move out of a safe position cannot capture. Let r be safe for (u_1, u_2) and let the cops move to (u_1', u_2') with $u_i' \in N[u_i]$. If $u_i' = r$ then either $u_i' = u_i = r$, contradicting $r \notin \{u_1, u_2\}$, or $u_i \sim u_i' = r$, contradicting that r is non-adjacent to u_i.

(B) Safety can be restored. Let $r = (x, y)$ be safe for the cop positions before their move, and let the cops now stand at $u_1' = (p, q)$ and $u_2' = (s, t)$; by (A), $r \notin \{u_1', u_2'\}$. We exhibit $r' \in N[r]$ safe for (u_1', u_2'); note that $N[r]$ consists of r together with all cells (u, v) with $u \neq x$ and $v \neq y$. Say a cop *threatens* r if it is adjacent to r, i.e. u_1' threatens r exactly when $p \neq x$ and $q \neq y$.

Case 0, neither cop threatens r.: Then r itself is safe for the new positions; the robber passes, $r' = r$.

Case 1a, both threaten and $p = s$.: Take $r' = (p, v)$ with $v \notin \{y, q, t\}$, possible since $m \geq 4 > |\{y, q, t\}|$. Then r' is in row p, so it shares a line with both cops; $p \neq x$ and $v \neq y$, so $r' \in N[r]$; and r' is neither (p, q) nor $(s, t) = (p, t)$.

Case 1b, both threaten, $p \neq s$ and $q \neq t$.: Take $r' = (p, t)$: it shares row p with u_1' and column t with u_2'; $p \neq x$ and $t \neq y$; and $r' \neq (p, q)$ as $t \neq q$, $r' \neq (s, t)$ as $p \neq s$.

Case 1c, both threaten, $p \neq s$ and $q = t$.: Take $r' = (u, q)$ with $u \notin \{x, p, s\}$, possible since $m \geq 4 > |\{x, p, s\}|$. Then r' is in column $q = t$, so it shares a line with both cops; $u \neq x$ and $q \neq y$; and r' is neither (p, q) nor (s, q).

Cases 2a and 2b, exactly one cop threatens r.: Relabel so that $u_1' = (p, q)$ threatens r and $u_2' = (s, t)$ does not, so $s = x$ or $t = y$; not both, since $(s, t) = (x, y) = r$ is excluded by (A).

Case 2a, $s = x$ and $t \neq y$.: Any $r' = (u, v) \in N[r] \setminus \{r\}$ has $u \neq x = s$, so sharing a line with u_2' forces $v = t$; and $t \neq y$ is given. If $t \neq q$ take $r' = (p, t)$, which shares row p with u_1' and column t with u_2', and differs from (p, q) and from (x, t). If $t = q$ take $r' = (u, t)$ with $u \notin \{x, p\}$, possible since $m \geq 3$; then r' shares column $t = q$ with u_1' and column t with u_2', and equals neither.

Case 2b, $t = y$ and $s \neq x$.: Symmetrically any $r' = (u, v) \in N[r] \setminus \{r\}$ has $v \neq y = t$, so $u = s$. If $s \neq p$ take $r' = (s, q)$: it shares column q with u_1' and row s with u_2', and equals neither. If $s = p$ take $r' = (s, v)$ with $v \notin \{y, q\}$, possible since $m \geq 3$; then r' is in row $s = p$ with both cops and equals neither.

No case imposes any condition on *which* move the cops made, so the argument covers every cop strategy. By (C) the robber can start safe; by (A) no cop move ever captures him; by (B) he restores safety after each cop move. Hence he survives forever and $\mathrm{c}(G_m) \geq 3$. Case 0 uses the robber's option to pass, which is part of the convention of [3]. $\square$

Theorem 4. *G is a connected perfect graph with $\alpha(G) = 4$ and $\mathrm{c}(G) = 3$. Consequently $f(4) = 3$: the bound $\mathrm{c}(G) < \alpha(G)$ of [3] for connected perfect graphs with $\alpha \geq 4$ is tight at $\alpha = 4$.*

Proof. Lemma 1 gives connectivity, perfection and $\alpha = 4$; Lemmas 2 and 3 give $3 \leq \mathrm{c}(G) \leq 3$. So $f(4) \geq 3$. The corollary of [3] quoted in §1 gives $f(4) \leq 3$. $\square$

Only the instance $k = 4$ of Problem 5.2 is settled here; no value $f(k)$ with $k \neq 4$ is determined.

3. Attribution

The graph G and the value $\mathrm{c}(G) = 3$ are already in print. Ahirwar, Bonato, Gittins, Huang, Marbach and Zaidman [1] study Cops and Robbers on *Latin square graphs*: for an $m \times m$ Latin square L, the graph $G(L)$ has the m^2 cells as vertices, two distinct cells being adjacent when they share a row, share a column, or carry the same symbol. They record that "by directly checking, the cop number of a latin square of order 1 or 2 is 1, order 3 is 2, and order 4 is 3".

Our G is such a graph. Let L be the Cayley table of the Klein four-group, $L(i, j) = i \oplus j$ on $\{0, 1, 2, 3\}$ with $\oplus$ bitwise exclusive or, and index the cells of $G(L)$ as $4i + j$, $0 \le i, j \le 3$. Then the permutation

$$\varphi = (0,\ 5,\ 10,\ 15,\ 11,\ 14,\ 1,\ 4,\ 13,\ 8,\ 7,\ 2,\ 6,\ 3,\ 12,\ 9),$$

read as $\varphi(k)$ for $k = 0, \dots, 15$ with vertex k of G being the cell $(1+\lfloor k/4 \rfloor,\ 1+(k \bmod 4))$, is an isomorphism $G(L) \to G$; a reader can check the $\binom{16}{2} = 120$ pairs directly. So the object of Theorem 4 is the order-4 Latin square graph of [1], and the integer 3 is theirs. What perfection and $\alpha(G) = 4$ add is that this graph is a witness for $f(4)$.

4. Verification

Every claim above is checkable by hand on the objects printed in §2 and §3. A program accompanies the note, `verify.py`, with its recorded output in `verify.output.txt`; it uses exact integer and bitmask arithmetic and no third-party package. It reads the `graph6` string, the adjacency rule, the broken diagonals, the dominating set, the six-case strategy and the permutation φ as printed here; it recomputes α, θ and the domination number by exact search; it certifies perfection twice, once by enumerating all 32,192 odd vertex subsets of size at least 5 for an induced odd hole or antihole and once by checking that the complement is $L(K_{4,4})$; it replays Lemma 3 over all 576 safe states and all 57,600 cop moves out of them, in the case-by-case form printed above, and independently decides the two-cop game by backward induction over all $16^3 \cdot 2$ states; and it verifies φ. Its recorded run reports

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VERDICT: ALL 37 CHECKS PASS
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No three-cop solver was run: $\mathrm{c}(G) \le 3$ rests on the printed dominating set, and independently on the corollary of [3] quoted in §1.

References

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[3] A. Clow and I. Zaguia, *Cops and Robbers, clique covers, and induced cycles*, Discrete Math. **349** (2026), 115252. doi:10.1016/j.disc.2026.115252; arXiv:2507.14321v1. Problem 5.2 and the corollary quoted in §1 are reproduced in full above from the arXiv source, version v1, which is the text we read; the numbering of the journal text may differ.

[4] L. Lovász, *Normal hypergraphs and the perfect graph conjecture*, Discrete Math. **2** (1972), no. 3, 253–267. doi:10.1016/0012-365X(72)90006-4.